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Data Analytics Pipeline Final Report

Project Overview and Goals

This project will mainly be focused on creating an earthquake data analytics pipeline using Google Cloud Platform (GCP) services. More specifically, data will be pulled from a dataset, stored, and processed to perform analytics to generate insights or results, simulating a real-world data workflow. In this case, this project will focus on earthquakes and the data collected during earthquakes, in an effort to potentially predict the magnitude of unseen earthquake instances. This will be done by creating a machine learning model using GCP services.

It should be noted, however, that this project is mainly to gain more experience in the data analytic pipeline and how it should flow, rather than to seriously predict earthquake magnitudes. In real life, scientists have never been able to predict a major earthquake, meaning they are unable to define the date, location, and magnitude of an earthquake. Instead, they can only calculate the probability that a significant earthquake will occur in specific areas within a certain number of years (“Can you predict earthquakes?”).

Dataset Description

The dataset used for this project is the “[Global Earthquake History: 1M+ Since 1900](#)” dataset with (currently) 1,057,653 rows, and new rows are being added. The dataset is a structured dataset from Kaggle, and records significant earthquakes (2.5+) since 1900, recorded by the United States Geological Survey (USGS). The dataset has 23 different columns representing different values, including the location of the earthquake, time, depth, magnitude, etc.

Description of Pipeline Workflow and Services Used

The pipeline workflow for this project mainly followed the stages of data ingestion, data storage, data processing, data warehouse, building a machine learning model, and visualization, following the core components.

Specifically, for this project, the pipeline first began with ingesting and storing the data. This involves downloading a CSV file of the dataset from Kaggle and storing the raw data file in Google Cloud Storage. After being stored, BigQuery was used to clean and normalize values in the dataset, as well as to derive new columns, as further explained below. A machine learning model, more specifically a boosted tree regressor, was then built based on the newly cleaned dataset through BigQuery. Afterwards,

different insights based on the data and the performance of the model were organized in Looker Studio.

However, after presenting the results of the model and being given feedback, certain tweaks were made to the machine learning model stage, wherein the model was retrained without certain features, before eventually updating the visualization with the new data.

To summarize, the core components of the data pipeline workflow were followed, while using Google Cloud Storage, BigQuery, and LookerStudio in the process

Data Processing Steps

Certain data processing steps taken to clean the data include using BigQuery to remove any rows with missing (NULL) values to ensure the model would run on complete data. Certain columns were then normalized to ensure consistent formatting (as there was inconsistency in capitalization), as well as focusing all data in the dataset solely on earthquakes (as events such as explosions are also included in the dataset). Further, additional columns were derived based on values in the dataset, including a new location column, the season the earthquake occurred, and the hemisphere the earthquake occurred in. This cleaned dataset was saved as a table in BigQuery, with the query used for this step displayed below:

```

1 CREATE OR REPLACE TABLE `cpsc482-project.earthquake_dataset.clean_earthquake_table`
2 AS
3     SELECT id,
4            time,
5            year,
6            month,
7            day_of_year,
8            hour,
9            ST_GEOGPOINT(longitude, latitude) AS location,
10           depth,
11           mag,
12           UPPER(TRIM(magType)) AS magType,
13           place,
14           LOWER(TRIM(type)) AS type,
15           CASE
16             WHEN tsunami = 1 THEN TRUE
17             ELSE FALSE
18           END AS tsunami,
19           mag_category,
20           depth_category,
21           nst,
22           gap,
23           dmin,
24           rms,
25           CASE
26             WHEN month IN (6,7,8) THEN 'Summer'
27             WHEN month IN (9, 10, 11) THEN 'Autumn'
28             WHEN month IN (12, 1, 2) THEN 'Winter'
29             ELSE 'Spring'
30           END AS season,
31           CASE
32             WHEN latitude >= 0 THEN 'Northern'
33             ELSE 'Southern'
34           END AS hemisphere,
35           year + (CAST(day_of_year AS FLOAT64)/365.0) AS decimal_year
36 FROM `cpsc482-project.earthquake_dataset.earthquake_table`
37 WHERE LOWER(TRIM(type)) = "earthquake" AND
38        depth IS NOT NULL AND
39        depth_category IS NOT NULL AND
40        nst IS NOT NULL AND
41        gap IS NOT NULL AND
42        dmin IS NOT NULL AND
43        rms IS NOT NULL

```

A machine learning model, specifically a boosted tree regressor, was then built using the cleaned dataset to predict magnitude.

Results Obtained

From this pipeline, four tables were produced, along with 2 models. The tables produced include the raw earthquake table, a clean table with the removed NULL values, as well as two tables combining the clean earthquake table with the different predictions produced by the models.

As briefly mentioned, two different machine learning models were built for this project. The initial one used nearly all features in the original dataset to predict magnitude, including a feature called “mag_category,” which holds a value based on the

magnitude itself. The second one does not use “mag_category” when predicting magnitude due to its dependence on magnitude itself. As such, this pipeline produced two different models and four different tables.

Additionally, both models were evaluated, with the model that included the mag_category feature returning the following values:

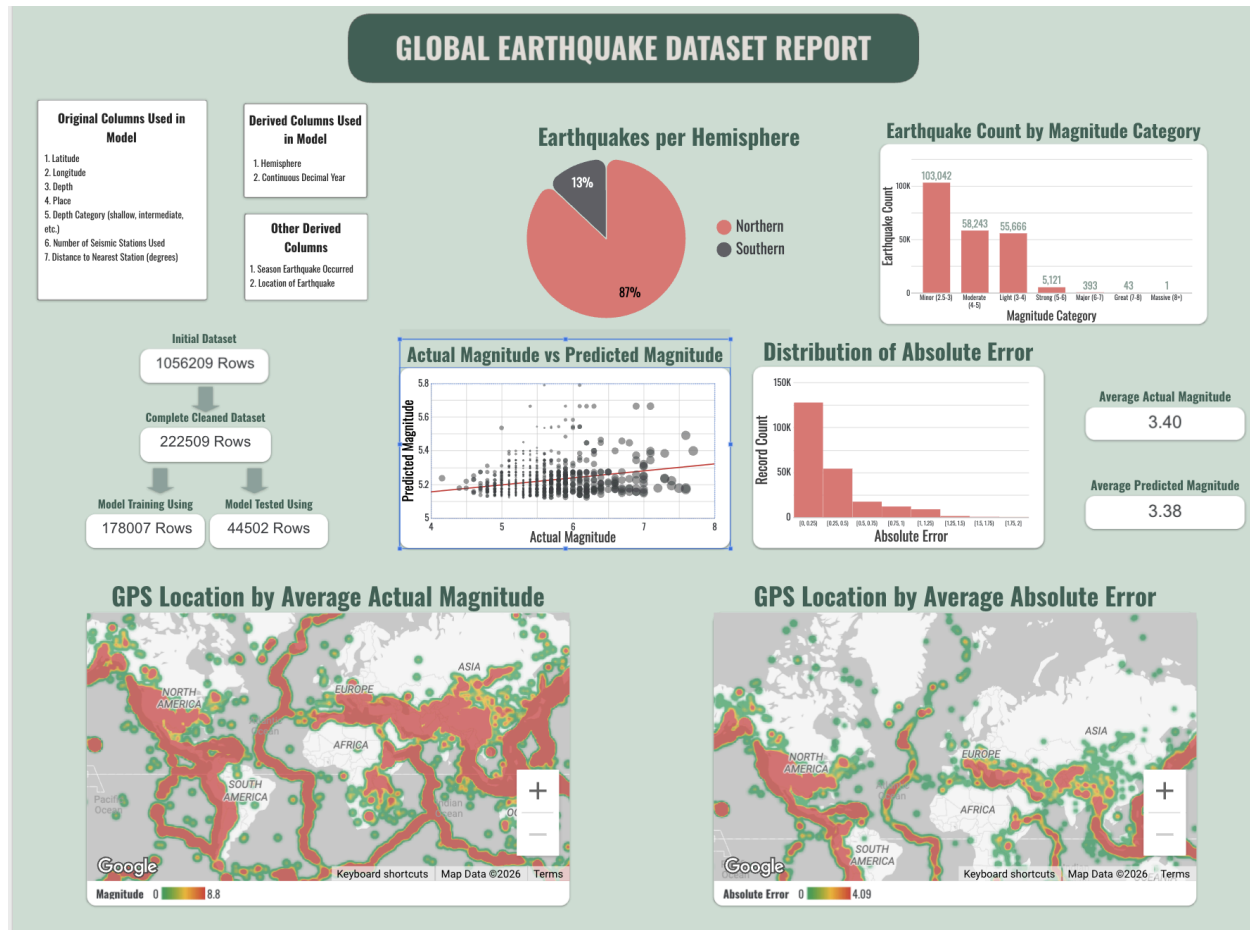
mean_absolute_error	mean_squared_error	root_mean_squared_error	r2_score
0.1456	0.0339	0.1841	0.9478

This model performed noticeably better than the model that didn’t include mag_category as a feature, returning the following values:

Row	mean_absolute_error	mean_squared_error	root_mean_squared_error	r2_score
1	0.2732	0.1465	0.3828	0.7706

Visualizations/Dashboard

An [interactive Looker Studio report](#) was also compiled to present any findings from the dataset and analysis, connecting directly to the tables in BigQuery. This report shows the distribution of earthquakes based on their hemisphere location, the distribution of the different levels of earthquakes, a heat map displaying the standard error, representing areas the model did not perform as well on, and much more. A full screenshot of the report is displayed below:



Challenges Encountered and how they were Resolved

One challenge or misunderstanding encountered during this pipeline was a noticeable spike in average magnitude over time that would appear in graphs in the visualization stage. As this was likely due to recent minor earthquakes simply not being added to the USGS database, the spike was likely not a sign of increasing magnitude over time. However, this could also have caused misunderstandings in others who view the data, and as such, graphs that show the average magnitude over time were removed to prevent any potential misunderstandings.

Lesson Learned/Overview of Learning

As previously mentioned, this project was not meant to produce a serious model that would accurately predict the magnitude of earthquakes, as scientists are unable to do this. However, this project did provide an important experience into how the data analytics pipeline works, how to work alongside and incorporate feedback, and how to visualize patterns in data in such a way that it would be intuitive for others to view using tools such as Looker Studio.

Potential Next Steps

Plans that could further improve this model could include transforming the pipeline to account for streaming data. This could allow dynamic updates as new data arrives from USGS. Further, allowing missing values to be used in the models could increase the number of training and testing datasets to the original number of rows.

Citations

“Can You Predict Earthquakes?” *USGS*,
www.usgs.gov/faqs/can-you-predict-earthquakes. Accessed 10 May 2026.